Inter-operator OSS interface for delivering pan-European ATM VP service

D W Ranasinghe, C C Smith, C B Hatch, S C Beaumont and P Kristiansen

The provision of ATM virtual paths across multiple network domains requires interaction between the operational support systems (OSS) used by the different network operators to manage those networks. This paper describes the interfaces needed between such OSS to support performance and accounting management. It also describes an approach to ensuring those interfaces are secure. The results may be suitable for operational deployment and applicable with further development for other ATM services such as virtual circuits and switched services.

1. Introduction

If ATM technology is to succeed outside the LAN market it must support customers of one operator in communicating with customers of another operator, and it must allow operators to offer service in geographic areas where they do not have their own network. Both of these criteria require the ability for an operator to create and manage an ATM connection across another operator's network. This paper describes an investigation into the interfaces that will be required between the OSS of different providers if they are to co-operate in offering pan-European ATM virtual path (VP) services.

The Telecommunications Management Network (TMN) architecture [1] defines basic concepts to describe the interfaces between operational support systems (OSS) and networks, and between OSS. It also defines the term 'X-interface' to describe the interface between OSS owned by different operators.

This paper focuses on the development of specifications for the functional area of performance and accounting management as applied to ATM VPs. A VP identifies connections which share a common path and receive the same quality of service (QoS). Performance management is based on monitoring QoS parameters. Accounting management is based on measurements of usage of the network resources. The investigation has been carried out as part of a collaborative EURESCOM project [2] and has been supported by a pan-European TMN Laboratory jointly created by the project partners.

Operational deployment of an X-interface would need to consider security measures to prevent unauthorised use of the interface. This paper will present a solution to this problem that has been investigated and prototyped.

For the specific case of ATM network management the X-interface is generally referred to as the 'Xcoop' interface in the remainder of this paper. Furthermore, telecommunications operators are generally referred to as public network operators (PNOs), in order to maintain alignment with current managed object descriptions in the existing standards. No assumptions are made concerning PNO policies for the availability or allocation of resources required to provide ATM VP services.

In addition to performance, accounting and security functions, other related network management functions involving use of an Xcoop interface are:

- configuration management of ATM VP connections,
- fault management (or alarm reporting) of ATM VP connections.

These management functions are described in established ETSI standards [3, 4]. The developments reported in this paper are partially aimed at creating additional ETSI standards to augment those already defined [3, 4], and consequently provide a technical basis for encouraging the availability of improved vendor products, suitable for the automated management of ATM-based networks and services.

The remaining sections in this paper are structured as follows.

Section 2 discusses the organisational models that have been considered for the X-interface. Sections 3 and 4 describe the Xcoop interface developments in the performance and accounting management areas. Section 5 describes an approach to ensuring the security of the X-interface.

2. Organisational models

Agreements between network operators are required so that traffic needing to be supported by more than one network operator can be processed according to clearly agreed, and contractually binding, terms and conditions. The generic term used for these agreements is 'service level agreement'. Such agreements are clearly needed to cover the traffic levels and technical parameters associated with the provision of service at the network-to-network interface. However, there are other agreements that are needed to take account of the management of interconnected networks.

Interconnect agreements need to be defined in terms of some organisational relationship between network operators. Historically network operators have adopted a cascade mode of organisational relationship for most services, as shown in Fig 1.

This model uses the service user and service provider relationship such that with *n* network operators providing a service then the *n*th operator takes a user role and the (*n*+1)th operator a provider role. It is self-evident that the *n*th operator is only aware of the (*n*+1)th operator and does not 'see' the (*n*+2) operator, etc. The primary characteristic of this model is that there is no single operator in an overall controlling role. This means that when a customer-facing operator needs to determine where a failure has occurred or where performance has been degraded, outside its jurisdiction, it is only the 'next' operator in the cascade that can be requested for information. Equally when a network operator detects a problem in the provision of service to another network operator a service degradation report contractually need only be sent to the up-stream network operator. The cascade model has been used for many years and will continue to be used. However, there are occasions where a different organisational model may be more applicable.

ETSI have defined the 'star organisational' model for environments involving a close grouping of network operators. This is shown in Fig 2. The primary characteristic of this model is that one network operator takes the role of co-ordination and control for the provision and administration of a service involving multiple network operators. It is important to appreciate that any operator within the group can take this role for those customers connected to its network. A controlling operator thus acts in a user role while all the other operators in the group, who have been selected to support a service by a controlling

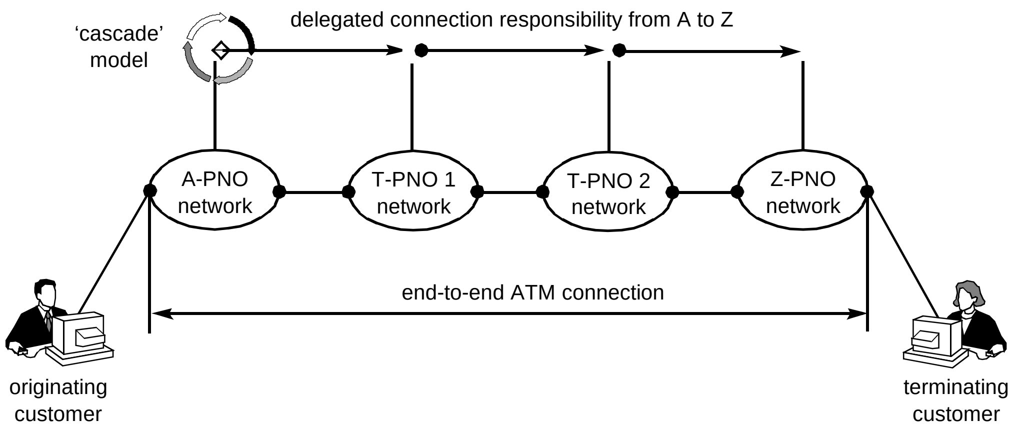


Fig 1 The cascade organisational model.

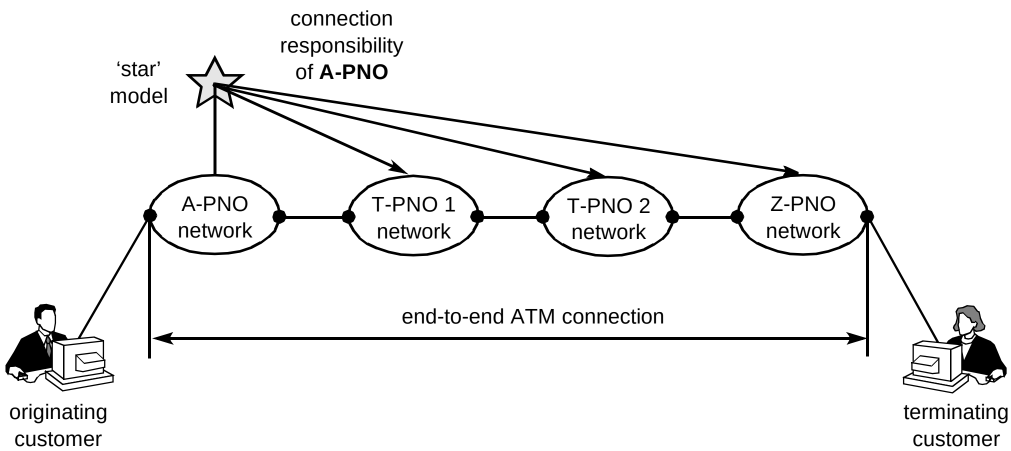


Fig 2 The star organisational model.

operator, play a provider role. A service is set up across these network operators by the controlling operator sequentially requesting each of the operators to support service between a network ingress point and network egress point, the egress point of one operator being connected by a transmission network to the ingress point of another operator. During operational service any operator detecting a failure or degradation of service, reports this to the controlling network operator, who is therefore aware of a problem and the identity of the network operator in difficulty.

3. Performance management

3.1 Overview

Before considering the EURESCOM performance studies, it is necessary to note that ATM is part of a layered transmission and switching architecture (see Fig 3).

In this model, which is realised in many commercial products, the ATM service has two layers — the ATM layer and the ATM adaptation layer. Higher layers from the perspective of the ATM service represent the user. Users expect the ATM service to transfer their information (voice, video, data, etc) reliably between geographically diverse locations. Users may also expect various degrees of reliable transfer, e.g. a very high level of assurance that information is delivered correctly or some lesser requirement. Where a very high level of assurance is required, the ATM service may need to use retransmission techniques to ensure information delivered to a user is correct. In other cases a user may be able to put up with a certain level of incorrectly received information because the user has its own recovery capabilities.

3.1.1 Network aspects of ATM

ATM is a cell-based switching and transmission technology using ATM layer cells of 53 bytes, which are partitioned into a 5-byte header field and 48-byte payload. The header is used for cell routeing, cell priority indication, operations administration and management (OAM) activity and may be used for flow control. However, flow control is only used between a so-called end user, such as a customer or private network, at a user-to-network interface (UNI). Between switches in the public or inter-carrier network, flow control is not used, although cell priority is used. This interface is the network-to-network interface (NNI).

The payload carries user information or perhaps more precisely information applicable to the layers above the ATM layer. This includes ATM adaptation layer information, which permits adaptation layer entities to communicate. Adaptation layer entities will exist at the ends of an ATM connection to interface with the user entities. Within the public network therefore, each switching node may typically only be concerned with the ATM layer.

3.1.2 Performance implications

It should be appreciated that from the perspective of a user the performance of the ATM layer may not necessarily directly impact the performance as seen by that user. This is because a certain level of ATM layer degradation may be removed by the layers above. The relationship between ATM layer performance and the performance as seen by a user must be appreciated and borne in mind when reading the remainder of this paper. It is apparent and obvious from this dependency relationship that the ATM layer must operate within specific performance targets.

Therefore the objective for a management system concerned with ATM layer performance is to ensure that performance is either within the target range or is known to be degraded and outside of that range. The action taken when performance is degraded is dependent on a defined recovery policy. Such a policy may require the use of alternative ATM paths by initiating protection switching at the ATM layer or to wait until performance has returned to its target range and once again to resume service. Where protection switching exists the user may notice a short break whereas in the latter case the outage will be dependent on the time taken for performance to return to normal.

EURESCOM studies were concerned with ATM layer performance for semi-permanent VPs and not the overall QoS provided by the ATM service as seen by a user. This statement needs a further qualification because the performance is actually the performance between an ingress point in a network operator’s environment and egress point in another network operator’s environment — in other words the ‘public’ network part of an ATM connection as shown in Fig 4.

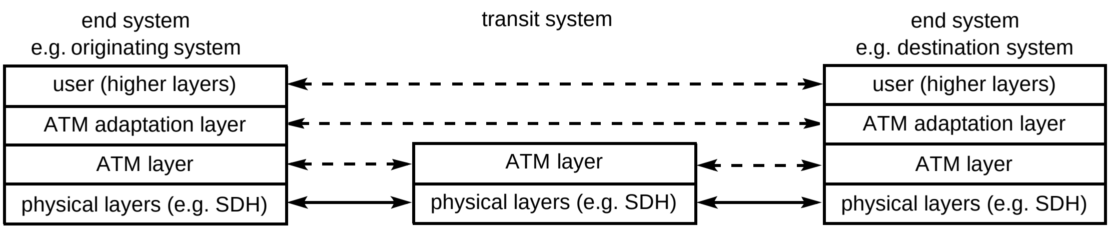


Fig 3 ATM layer’s relationship to higher and lower layers.

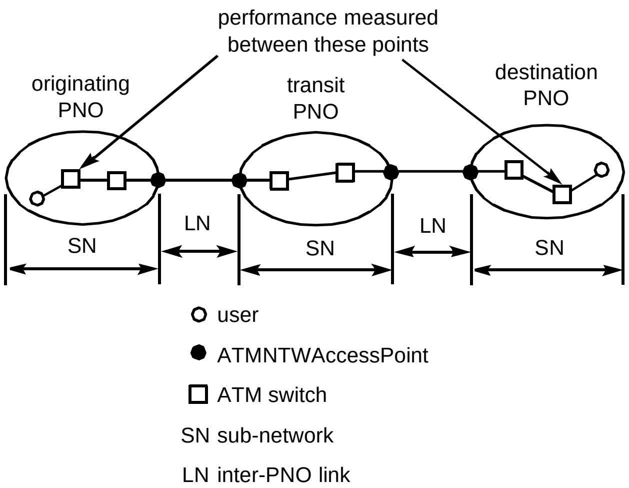


Fig 4 Semi-permanent VP performance.

3.2 Performance measurements

3.2.1 Measurement requirements

ITU-T I.356 [5] defines the performance parameters for the ATM layer connections together with values for these parameters and reference international connections. Consideration of the value range of the parameters was not within the scope of the EURESCOM studies and it was merely assumed that all these parameters were required to be measured for operational purposes. Parameters defined in ITU-T I.356 [5] are:

- cell transfer delay (CTD),
- cell delay variation (CDV),
- cell loss ratio (CLR),
- cell error rate (CER),
- cell mis-insertion ratio (CMR),
- severely errored cell block ratio (SECBR).

The meaning of these parameters will be generally obvious to the reader but the formal definition together with example ways of measuring these parameters is contained in ITU-T Recommendation I.356 [5]. Parameters are quoted as an upper bound on a probability distribution, but, for the EURESCOM studies, a threshold value was used as practical for operational purposes. Thus if ATM layer performance degraded beyond a threshold value a notification of this would be sent to a management system. This is the primary measurement requirement.

These parameters may be used for ‘in-service’ assessment of VP performance using special OAM cells or ‘out-of-service’ measurement using specialist test equipment. The requirement for the EURESCOM study is for ‘in-service’ measurement.

3.2.2 PNO relationships

Figure 5 shows a pan-European routeing involving an originating PNO, transit PNO and destination PNO. Each of these PNOs has a management system, which, because a TMN environment is being considered, is shown as a TMN operations system (OS). For management purposes the relationship between these PNOs is compliant with the ETSI star organisational model. This means that one PNO is responsible for setting up the VP as explained earlier. This PNO, which is shown as the originating PNO in Fig 5, is also the PNO that faces the user. Because the other PNOs have been requested to provide a VP across their network by the originating PNO, the relationship they have with the originating PNO is of a service provider to the originating PNO user. They must therefore report any problem (fault,

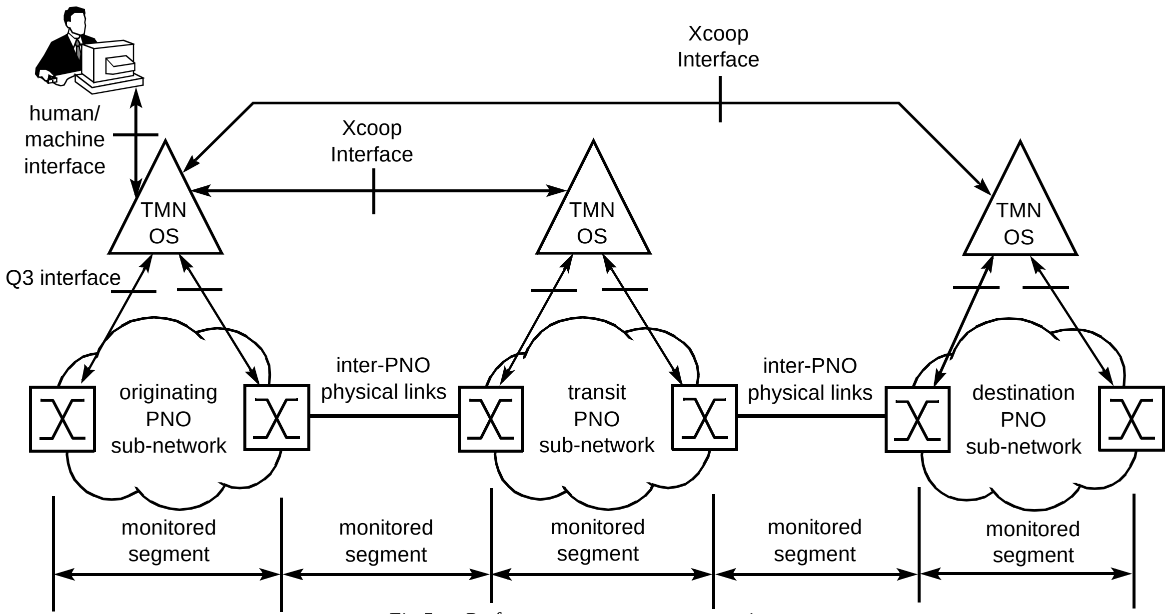


Fig 5 Performance management scenario.

performance, etc) in providing the VP service to the originating PNO.

Thus the requirement is for each PNO to be able to measure performance using the parameters detailed above. Where a PNO detects that performance has been degraded because one or more parameter thresholds have been crossed, it will report this to the originating PNO. Sometime later the PNO reporting the degradation may determine that performance has returned to normal for any or all of the parameters and this will also be reported to the originating PNO. The threshold process is shown in Fig 6.

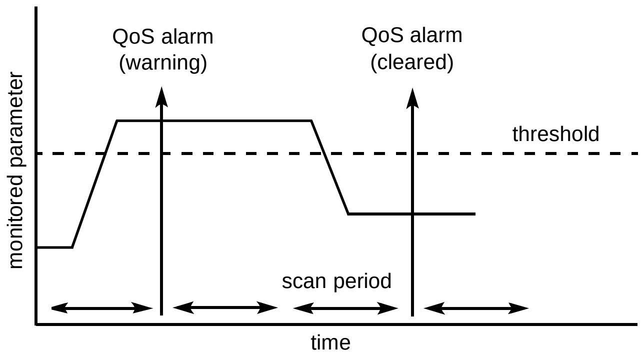


Fig 6 QoS alarm and QoS alarm clearance notification.

Each of the parameter values measured will be compared with its own threshold, but only once per scan period. The scan period can be varied as necessary for operational purposes. Therefore if during a scan period a threshold is crossed in the upward direction, it can only be reported as returning to normal at a subsequent scan period. The actual measurement of performance is an ATM switch function while the threshold for inter-PNO performance is an OS function.

3.2.3 VP partition into ATM segments

ITU-T I.610 [6] defines the way a VP may be partitioned for performance purposes. At the VP level there is an F4 cell flow for OAM purposes. This flow is in addition to the user cell flow and effectively partitions the user’s cells into blocks with an OAM cell at the start and end of a block. There are several types of OAM cell defined by ITU-T, including one defined for performance use — the OAM-PM cell. Under network management control an OAM cell may be introduced into a VP at an ingress point and extracted at an egress point. OAM-PM cells have a sequence number and a count of the user cells. This permits a receiving switch to determine two important performance-related aspects — whether an OAM-PM cell has been lost and the number of user cells in a block delineated by two OAM-PM cells. Using this and other information an ATM switch may assess the performance across the segment.

Two types of measurement and F4 flows may be used:

- end-to-end measurement (i.e. between user points in Fig 4),
- segment measurement.

Each of these F4 cell flows can co-exist. Therefore it is possible concurrently to carry out end-to-end performance assessment and segment assessment. Because the EURESCOM study was concerned only with that portion of a VP that starts in the originating PNO and terminates at the destination PNO, the end-to-end flow is not considered. The end-to-end flow is available for a user to carry out their own performance assessment over the whole VP.

In EURESCOM, the segments shown in Fig 5 were defined. These are:

- sub-network,
- inter-PNO physical links.

Each of these segments has an ingress point and egress point for OAM cells. In the case of the inter-PNO physical links these points are located in different PNOs. It is the responsibility of each PNO to control the actual insertion and extraction of OAM cells. However, the initiation of this insertion/extraction process for performance management may be initiated by the originating PNO.

The originating PNO will therefore see the performance of the public network part of the VP as a collection of segments and not as a single public network VP. This is consistent with the contractual agreement between the originating PNO and the other PNOs supporting the VP. Therefore, if any PNO is not providing an acceptable QoS, the originating PNO will be able to determine this. Knowing where performance is impaired provides the originating PNO with information about what recovery action is required in support of the contractual service agreement with the user, for example:

- initiate fast re-routeing to use an activated standby VP between originating and destination PNOs,
- initiate a new VP (a relatively slow process),
- wait until the PNO with the performance problem has returned to normal service.

There is another reason why the originating PNO needs to know which of the other PNOs is not providing a satisfactory level of performance. This is for accounting purposes (see the next section).

3.3 Xcoop inter-carrier messages

3.3.1 Control of performance monitoring

The time sequence diagram (Fig 7) shows an example of the message flows between the originating PNO and in this case a transit PNO. However, the example is true for any other PNO involved in the VP. This is because the VP is configured by the originating PNO who requests each PNO in turn to reserve a sub-network connection in accordance with the ETSI configuration standard. If a PNO supports performance management, when a ‘reserve sub-network connection’ request is received it will also prepare to carry out performance management. Some time later the originating PNO will request the transit PNO to start performance measurement. This will cause the T-PNO to allocate space in its management information base for the monitored results, but, as shown in Fig 7, will not cause monitoring to start. Requests to start performance monitoring are either directed to the sub-network or VP link (shown as inter-PNO physical links in Fig 5). However, monitoring of both VP link and sub-network connection is possible concurrently. Furthermore, it is possible to specify whether one or both directions of transmission are to be monitored.

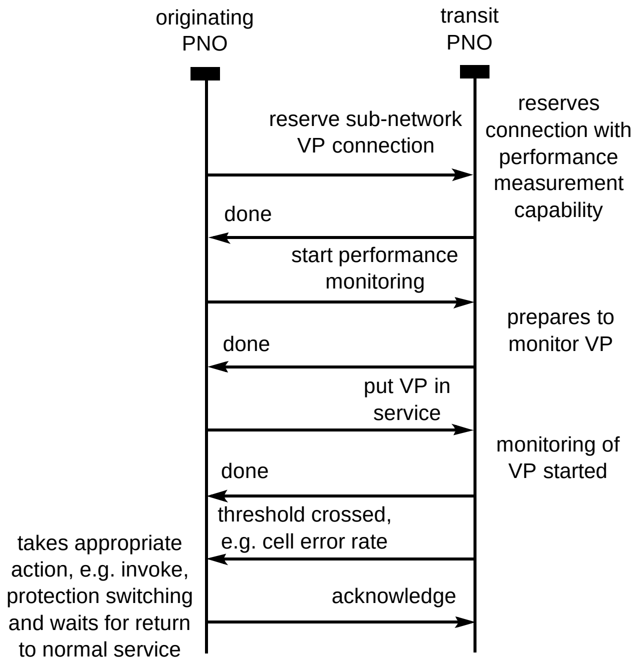


Fig 7 VP reservation and control of VP performance monitoring.

When the VP is put into service, which in TMN terms means being set to administrative state unlocked, performance monitoring will be started. If the originating PNO had not requested performance to be started, it may do so at any time provided the transit PNO still has resources allocated to the VP.

Performance monitoring may be stopped at any time by the originating PNO requesting the transit PNO to stop monitoring. Once stopped it is possible to restart monitoring. This provides the originating PNO with full control over the monitoring activity. It is, however, important to note that this control only applies to the Xcoop activity. Any internal performance monitoring that the transit PNO needs to carry out is not under the originating PNO control.

3.3.2 Performance reporting

While performance monitoring is active, the originating PNO may at any time request the current value of the performance parameters using the ‘read performance request’. In response to this request the transit PNO will return the current performance data. If the request is for all performance data, namely VP, sub-network and all performance parameters, this data will be returned. Alternatively the originating PNO may selectively request either sub-network or VP link and specific performance parameters. It is also possible to select one direction of transmission. This ‘read’ request is entirely independent of the ‘performance alarm report’ message described in the next paragraph.

If a parameter threshold has been crossed, the transit PNO reports this, using the ‘performance alarm report’ message, to the originating PNO. This event report includes the following information:

- identity of the VP,
- event time,
- event type (performance alarm),
- severity (warning or cleared),
- type of threshold crossed (e.g. cell error rate),
- observed value.

In addition to sending a ‘performance alarm report’, the transit-PNO will log the alarm notification locally. Over time a historical record of VP link and sub-network performance may be held in this log. The originating PNO can interrogate this log using a ‘performance alarm log inspection request’. This request can include a filter predicate to only select specific performance information of interest, e.g. list all cell error reports within a stated time window.

On receipt of a ‘performance alarm report’ the originating PNO will determine what action is needed to minimise service problems being suffered by the user. It will then initiate an appropriate recovery process. Recovery actions are outside the scope of the performance management study. However, as part of another

EURESCOM study, detailed proposals are being completed to enhance the ETSI configuration management standard with recovery capability. This includes ‘hot’ VP stand-by (called fast re-routeing) where a complete alternative VP can be used by the originating PNO.

3.3.3 Threshold setting

ITU-T I.356 [5] defines a hypothetical reference connection with end-to-end objective values (a sort of performance budget) for ATM performance parameters (i.e. CER, CTD, etc). Because a connection will involve at least two PNOs (originating and destination) and possibly one or more transit PNOs, there is a need to have rules for allocating a part of the end-to-end performance budget to PNO portions. ITU-T I.356 [5] does just this and contains rules for determining what part of the budget a PNO, who may be part of a 2, 3, 4 or more PNO chain, may be allocated. In the EURESCOM studies the PNO portion is partitioned into two parts — the sub-network and inter-PNO links. The rules in ITU-T [5] can be used to determine the overall allocation for a PNO, but also allow the flexibility for a PNO to allocate its ‘budget’ between a sub-network and inter-PNO link.

The originating PNO therefore need not, and currently does not in the EURESCOM Xcoop interface, request specific initial threshold values. Instead, as part of the reserve request, the originating PNO includes a QoS parameter. This is used for example by a transit PNO to obtain parameter values from a QoS table held by the transit PNO. The values in this table provide specific threshold values for the sub-network and inter-PNO links in the direction towards the destination PNO. Values are not held for inter-PNO links in the direction towards the originating PNO because this link is the responsibility of the originating PNO. This table is not included in the EURESCOM interface as a standard item but is a locally defined and managed entity. This allows a PNO to use a rule-based approach that can include the ITU-T allocation rules and other PNO specific rules to allocate target values based on the most efficient resource allocation and routeing in that PNO’s environment.

- Same value

  Initial threshold values for the VP link in the direction towards the originating PNO are obtained after the reservation stage and come from either the PNO in the direction towards the originating PNO or the originating PNO itself. For example, in Fig 8 the parameter values used by T-PNO#2 originate from T-PNO#1. Previously T-PNO#1 may have responded to an originating PNO reservation request with a link identifier that both T-PNO#2 and itself will use for identifying their common link connection. As part of this response T-PNO#1 provides parameter values for the link connection towards T-PNO#2. These parameters are received by T-PNO#2 in the request to start performance monitoring. Therefore when a request to start performance monitoring is received T-PNO#2 has sub-network and all inter-PNO link threshold values to use in the monitoring activity.

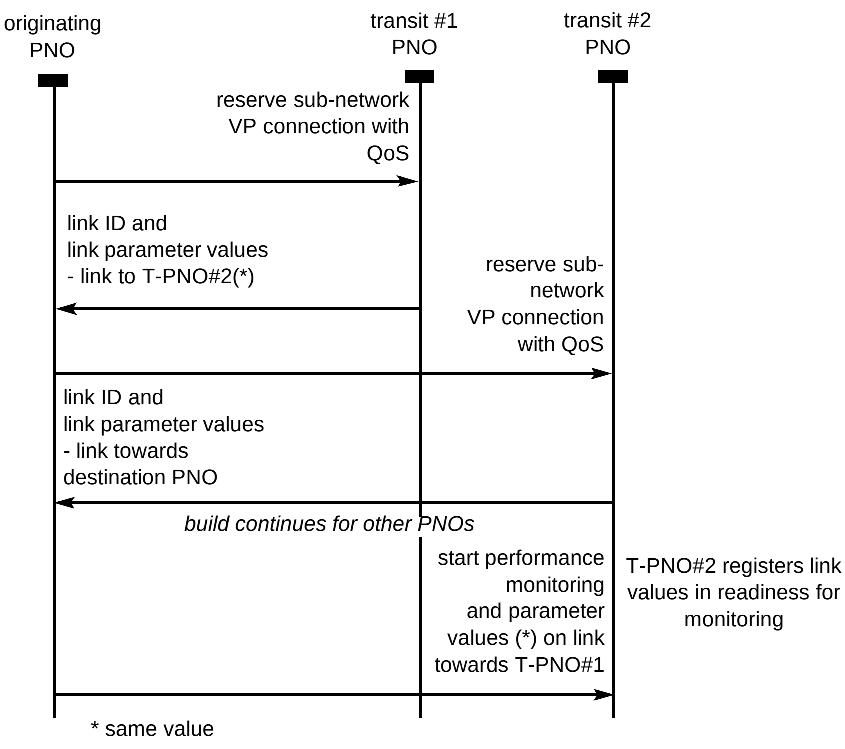


Fig 8 Initialisation of VP link performance parameters.

- Additional thresholds

  In addition to the 'regular' or initial thresholds, the originating PNO may request a PNO to apply 'additional' threshold monitoring to one or more of the performance parameters. For example, if performance monitoring is taking place, an originating PNO may request 'additional' threshold value monitoring by providing the threshold values to be used. The PNO will apply these thresholds in addition to the regular thresholds. This allows the originating PNO to set thresholds either side of the regular thresholds. If the 'additional' threshold is set to be more stringent than the 'regular' threshold, an indication that performance is being degraded to potentially unacceptable levels could provide a warning message to the originating PNO with the regular threshold providing a critical or service-affecting message. Subsequent to setting additional threshold values, these values may be changed. The Xcoop interface definition includes rules to ensure that coherent reports are received by the originating PNO when additional thresholds are changed.

3.4 Information model

3.4.1 Management view

The Xcoop message interactions described in the previous section are realised as a TMN-based interface. This means that it uses CMIP ITU-T X.711 [7] as the carriage protocol and is specified as an object-oriented information model. The model is an extension of the ETSI standard configuration model. Figure 9 shows the managed object containment tree.

In interface terms the information model is the performance management view that an originating PNO has through an Xcoop interface of any PNO that can support a VP with performance monitoring capability. Figure 9 is the view for one PNO with one VP sub-network connection. However, in practice, it is likely there will be several VPs being managed, each with associated managed objects, as in Fig 10. The tinted managed objects are part of the ETSI configuration standard, with the others being developed by EURESCOM specifically for performance management.

3.4.2 Managed object relationships

Managed objects are defined using the TMN inheritance ITU-T X.720/722 [8, 9] method of specification. Figure 10 shows the inheritance relationships for performance management. Only those objects specific to performance have been included. The log-managed object is a standard ITU-T X.721 [10] object but the other objects were developed by EURESCOM. The alarmRecord will be contained in the subNetwork object and will include information on all VP related alarms.

The primary role of the various managed objects is as follows.

- PnoVpSubnetwork (not shown in Fig 9)

  This managed object is defined in the ETSI configuration management standard [1]. A PNO that supports the creation and management of a semi-permanent VP will include this managed object. The object will exist whether or not there are any VPs actually created. A VP reservation request received from an originating PNO is received by this managed object, which initiates the VP reservation process on receipt of the request. If available sub-network resources are allocated and a PnoVpSubnetwork Connection managed object created. In addition, and only if performance management is supported, BidirectionalPerformanceMonitor managed objects will be created in the PNO's management information base. Initialisation is completed when a response has been sent to the originating PNO, confirming that the sub-network is reserved and performance capability has been allocated.

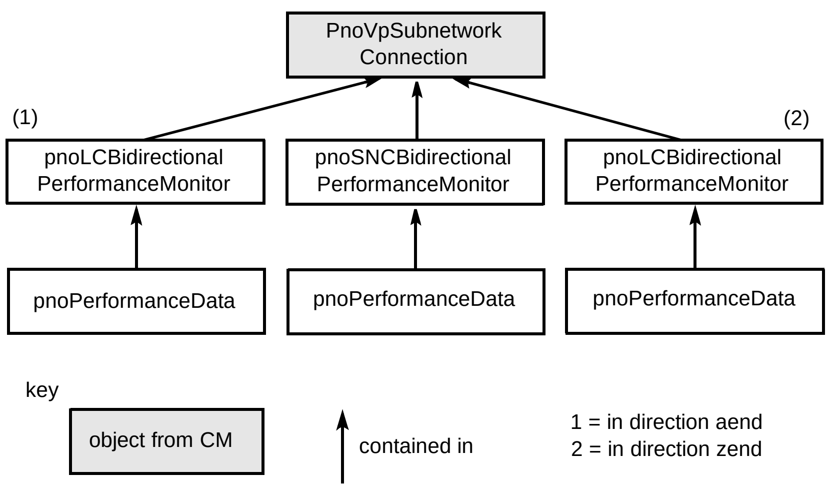


Fig 9 Containment tree for one sub-network.

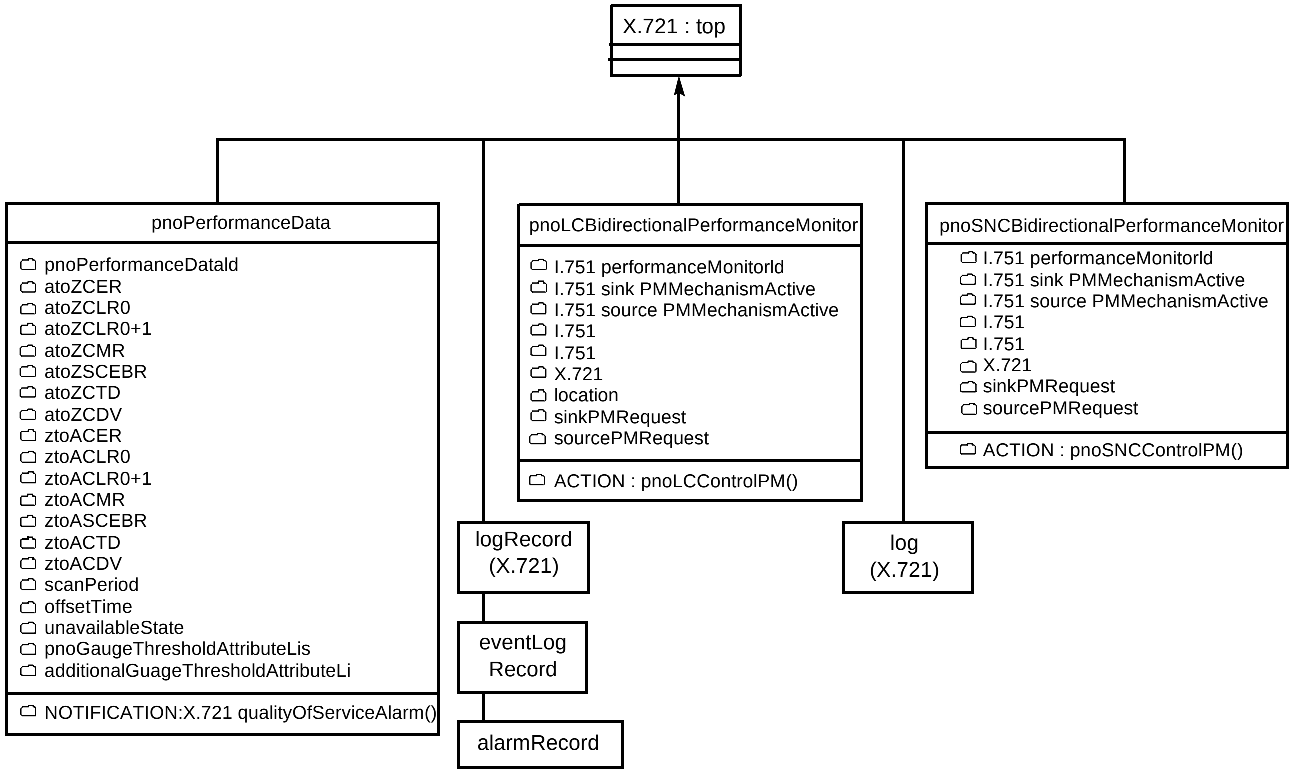


Fig 10 Managed object inheritance relationships.

- PnoVpSubnetworkConnection

This managed object is defined in the ETSI Configuration Management standard. It is created when a PnoVpSubNetwork-managed object receives a sub-network reservation request. A PnoVpSubNetworkConnection-managed object will exist as long as a PNO has reserved or is actually using its ATM resources for the requested VP. In performance management its only role is to act as a parent node in the naming structure so that all subordinate performance management managed objects are named relative to this object. The naming structure is important because it permits the remote originating PNO to address performance requests to those performance-managed objects associated with the target VP.

- PnoSNCBidirectionalPerformanceMonitor

Its primary role is to control performance monitoring for Xcoop purposes of an associated sub-network connection. This object receives a request to start or stop performance monitoring from the originating PNO. Performance monitoring can be requested to start for either or both forward and backward ATM cell-streams flowing between a sub-network ingress point and egress point on the same sub-network. These points are generally termed ‘sinks’ and ‘source’ network points. Clearly the ‘source’ in the forward direction is towards the originating PNO and that in the backward direction is nearest the destination PNO. A start request will result in the PNO initiating OAM-PM cells at the ingress point and extraction of those cells at an egress point. However, the VP must be in an administratively unlocked state, and, if this is not the case, performance monitoring will not actually be started until this state applies. The originating PNO may at any time send a request to this object to determine the current state of performance monitoring. A managed object of this class is created as a result of the PnoVpSubNetwork-managed object accepting a VP reservation request.

- PnoLCBidirectionalPerformanceMonitor

This object carries out the same role as the pnoSNCBidirectionalPerformanceMonitor but for inter-PNO links. There are two object instances of this class — one for the direction towards the originating PNO and one for the direction towards the destination PNO. Clearly these only control one end of an inter-PNO link, that terminates on the associated PNO. The other end is under the control of the ‘remote’ PNO. To initiate performance monitoring across a VP link connection, the originating PNO must send a message to pnoLCBidirectionalPerformanceMonitor objects in both PNOs. These are separate CMIP messages co-ordinated by the originating PNO.

- PnoPerformanceData

This object is created only when performance monitoring is started and deleted when monitoring is stopped. Creation and deletion activity occurs as a result of an internal PNO process that takes place when the associated BidirectionalPerformanceMonitor object receives a request to start or stop performance monitoring. Threshold values for ‘regular’ and ‘additional’ performance thresholds are held in this object for the inter-PNO link or sub-network as applicable. The inter-PNO link or sub-network role is determined by the parent object and is clear from Fig 9. These threshold values are held in a ‘gauge’ class of attribute, the behaviour of which carries out threshold testing. This means that, at a period set by a scan attribute, performance values, which originate from OAM-PM cell-based measurements carried out by ATM network elements (e.g. an ATM switch), will be compared with the threshold values held in these gauge attributes. If a threshold is crossed, a performance alarm notification is issued. Measured performance values are held in a number of attributes, for example, referring to Fig 10, the ztoACER attribute will hold the current measured values for cell error ratio measured for the monitored entity in the destination-to-originating PNO direction. An originating PNO may obtain the prevailing VP performance value, to temporal scanPeriod accuracy, from this object.

Performance measurements may be taken at a cadence specified by the originating PNO. This is possible because the value held in the scanPeriod attribute may be changed by the originating PNO. The offsetTime attribute records the difference between the current time and the time of the last update of performance parameters.

3.5 Operational exploitation

From an operational viewpoint the EURESCOM interface may well prove to be a practical solution for managing the performance of ATM semi-permanent VP connections, which traverse multiple PNO sub-networks. However, there are a couple of questions that need to be considered before operational deployment.

- Can the performance parameters stated in ITU-T I.356 [5] be measured accurately?
- Is overall performance assessed on a segment basis adequate for assessing overall VP performance?

3.5.1 Parameter measurement accuracy

One needs to recall that the EURESCOM project has developed a solution for ‘in-service’ ATM VP connections. In other words, while user cells are flowing this solution will provide an assessment of VP performance. Clearly there are ‘out-of-service’ measurements that can be carried out using specialist test equipment.

This implies that performance could be assessed by:

- in-service measurement only,
- out-of-service measurement only,
- use of both these techniques.

With out-of-service measurement, the complete cell-stream is available for performance measurement. There are two key points of note. Firstly, 100% of ATM cells are available for use in measurement although in commercially available measurement equipment there may well be a few housekeeping cells that could reduce this figure marginally below 100%. Nevertheless, in practice, virtually all the cells are being inspected. In this environment it is quite viable to accurately measure most parameters. The obvious limitation being that it estimates the performance of in-service VPs by an indirect method, i.e. a sample of VPs. One can question whether an out-of-service measurement that shows a sample of VPs to be operating within performance bounds will give the confidence that all in-service VPs are operating within an acceptable performance range. Clearly sample size is important to the resulting accuracy.

In-service measurement may be applied to 100% of semi-permanent VPs, but clearly this will be a cell-sample-based scheme because assessment is based on the use of OAM-PM cells. As these cells are few in number against user cells, in-service measurement is less accurate. This is recognised in ITU-T I.356 [5] because there are items for ‘further study’. For example, estimation of cell error ratio is noted to be ‘difficult’. Furthermore the method described by ITU-T, namely the use of OAM-PM cells, assumes that there are a small number of cells in a block, e.g. 200. An OAM-PM cell is placed at the front and end of a block. It may be that, provided there are few errors and large bursts of errors do not occur (noted by ITU-T I.356 [5]), practically accurate results will be obtained. What is not clear is at what error rates defined by ITU-T, operationally useful results will be obtained by the OAM-PM cell method.

There is also a question about the need to use all the performance parameters defined in I.356. This is an open question as are the actual values used in ITU-T I.356 [5] for reference ATM connections.

It may be that, with specific operationally based research study, the real issue for using the EURESCOM solution will be seen in terms of the parameter threshold values chosen; in these studies, they were not considered — the ITU-T values were assumed to be adequate. If one takes this view it would seem quite possible, that with a suitable choice of threshold values, there will be operationally accurate results obtained. This being so, operational decisions could be based on this method of performance assessment to determine when performance had degraded to a level that recovery procedures, such as the use of hot standby VPs, need to be invoked.

The major point arising from this brief consideration is that the performance assessed using the EURESCOM solution will not be as accurate as demanded in theory but may well be adequate for operational purposes.

3.5.2 Segment-based performance measurements

From a contractual aspect there is a need to ensure that the originating PNO is aware of the performance of each PNO supporting the VP for accounting purposes and other reasons. However, in addition to this, the originating PNO needs to have confidence that the overall VP, as provided to the user, is operating within acceptable performance bounds. The EURESCOM solution will not directly provide to the originating PNO the VP performance between the originating and destination PNOs. Instead the originating PNO will need to deduce performance from individual segment performance information. It would seem that provided threshold values for performance of each segment are chosen correctly then the multiple segment method of assessing overall VP performance will be adequate for operational reasons. However, this needs to be confirmed by operationally based studies.

As a result of the EURESCOM study it has become apparent that the method of segment definition defined by ITU-T has some practical limitations. Effectively this is that a segment cannot contain another segment. This means that, if a PNO has initiated a sub-network segment in support of the originating PNO's needs, it cannot initiate a segment within its sub-network on this same VP. Therefore if there is a need to carry out a test across an ATM switch on the VP (assuming there are several switches in the PNO's sub-network), this test cannot be carried out concurrently with the sub-network test. It is unclear if this is an operational limitation. If it is, the ITU-T segment definition needs to be enhanced to take account of this.

4. Accounting management

The development of a usage-based accounting management function for the Xcoop interface is taken to be both required and justified on the presumption that currently available fixed or subscription charging for ATM VP services will be increasingly unattractive in a developing competitive multi-provider market [11]. The proposition is that PNOs and customers would want to charge and be charged on the basis of:

- resource usage,
- market conditions,
- QoS provided,
- QoS guaranteed.

Accordingly, the need for a flexible, but accurate and verifiable, ATM usage-based accounting (and related charging) scheme seems compelling. This section considers the requirements and processes for such accounting, although the Xcoop interface specification remains to be formally developed.

4.1 Management of ATM VPs in the multi-operator environment

Figure 11 depicts the scenario where a (semi)-permanent VP has been configured across three network domains which are interconnected at the network (NNI) and network management levels (Xcoop).

Before explaining the details concerning the basis for information acquired and transferred to support usage-based accounting management, the location of 'measurement points' is described. Such measurement points for interconnected ATM VP services are identified as MPT or MPI in Fig 11 and are aligned with the definitions in ITU-T I.353 [12]. There is a general presumption that measurement points are located at interfaces with customer equipment or networks, and at network 'gateways'.

Specifically, MPT is defined as the usage measurement point at the interconnection with customer premises network/equipment (CPN/CPE), i.e. for connections originating across a UNI. The MPI measurement points are located at sub-network gateways. Accordingly, the locations of all measurement points are expected to coincide with PNO sub-network ingress and egress points for customer and/or PNO interconnections.

4.2 Reference model and resources for usage-based accounting management

Figure 12 provides more detail about the reference model and resources identified for usage-based accounting management in relation to the various functions of the Xcoop interface.

For the model shown in Fig 12, the roles of the respective PNOs are taken to align with the definitions in the ETSI specification [3], i.e. an originating operator (A-PNO) is connected to transit and destination operators (T and Z-PNOs) at the Xcoop network management level and at the NNI level, for the purpose of supporting end-to-end customer service delivery. For all PNOs the 'q_{any}' interfaces to the network elements are taken to be proprietary and are not discussed explicitly in this paper. Functions in the NNMS are defined as internal to any given PNO's processes, whereas functions in the respective INMSs are associated with inter-PNO processes. Further, the model in Fig 12 assumes that all PNOs have the same management functionality and interconnection resources and may interchange these roles (A, T, Z) according to

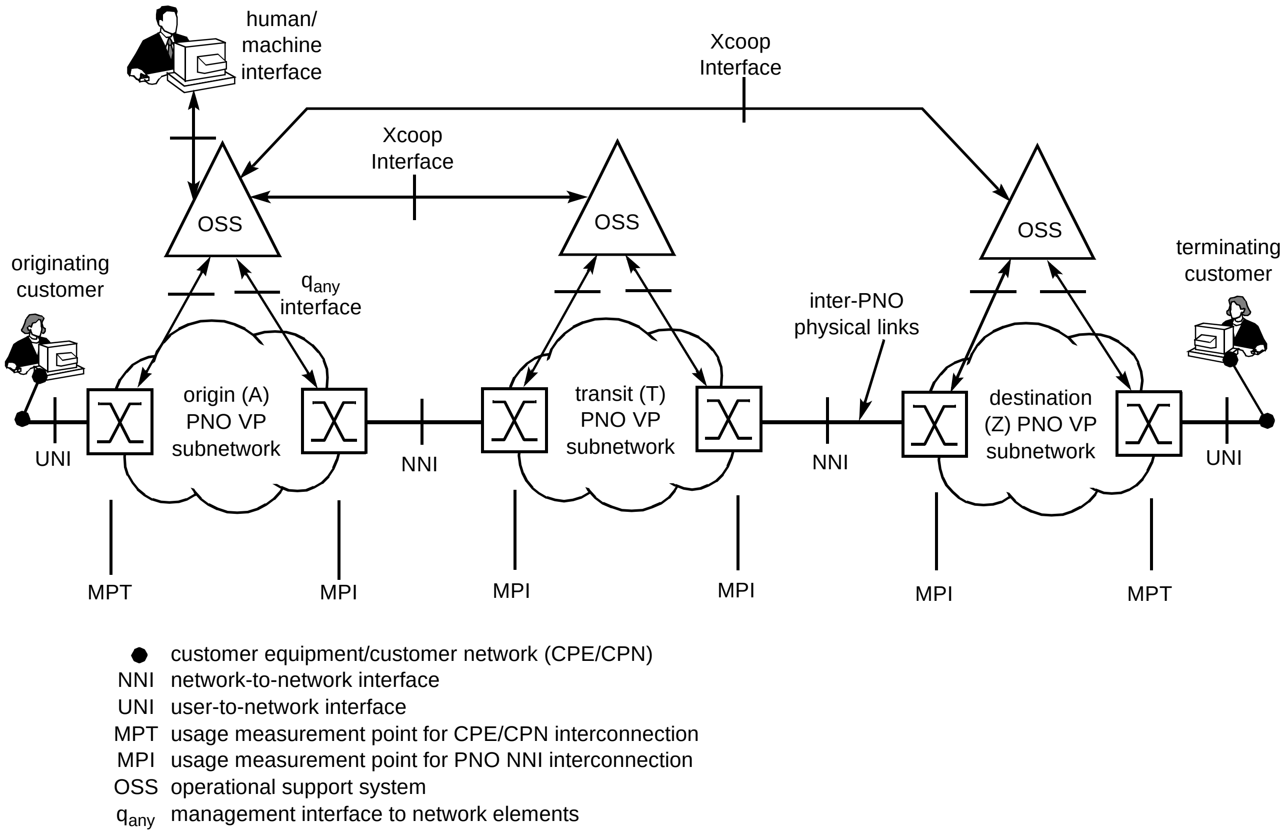


Fig 11 Accounting management scenario.

where any given ATM service is originated and terminated. A customer or service provider is also shown interconnected to the A-PNO, in particular with its charging and billing processes, to illustrate the functions associated with accounting management for acquisition of revenues.

4.3 *The standards basis for definition of usage-based accounting management*

The basis for usage-based accounting is taken to be a 'usage metering process' at the network elements associated with the MPI and MPT measurement points depicted in Fig 11. The usage metering process, depicted in Fig 12, is designed to generate usage metering records (UMRs), which may be modified by information input from performance monitoring processes. An aggregation of UMRs provides the connection detail record (CDR) for the ATM VP. Most significantly, the UMR data generation processes and the associated data collection, recording and reporting may be implemented in alignment with established standards [13—15]. A full specification of the Xcoop interface, based on these standards remains to be developed.

4.4 *Generation and exchange of X-CDRs*

For usage-based accounting management, CDRs should be generated whenever an 'accountable event' occurs which relates to the VP that is being monitored. Accountable events could include connection establishment, parameter changes, QoS degradation, unavailability, or at a fixed time interval trigger. The function defined as 'preprocessing' (see Fig 12) provides the CDR in the required data format. This, in turn, should be stored in a CDR log.

The CDR data format should contain:

- cell count information from the usage metering records, including cell loss priority, disruption duration and mean cell rate,
- customer connection information, e.g. connection reference, originating carrier ID,
- contract information, e.g. traffic descriptor type,
- traffic parameters, e.g. peak cell rate, sustainable cell rate, maximum burst size,
- extra services, e.g. bill type and bill information.

In general, the CDR should contain sufficient information to form the basis for charging a customer in a single-PNO environment, but needs to be augmented into an X-CDR for use in an Xcoop environment.

Figure 13 indicates the information which should be contained in the CDR and X-CDR data files. It seems

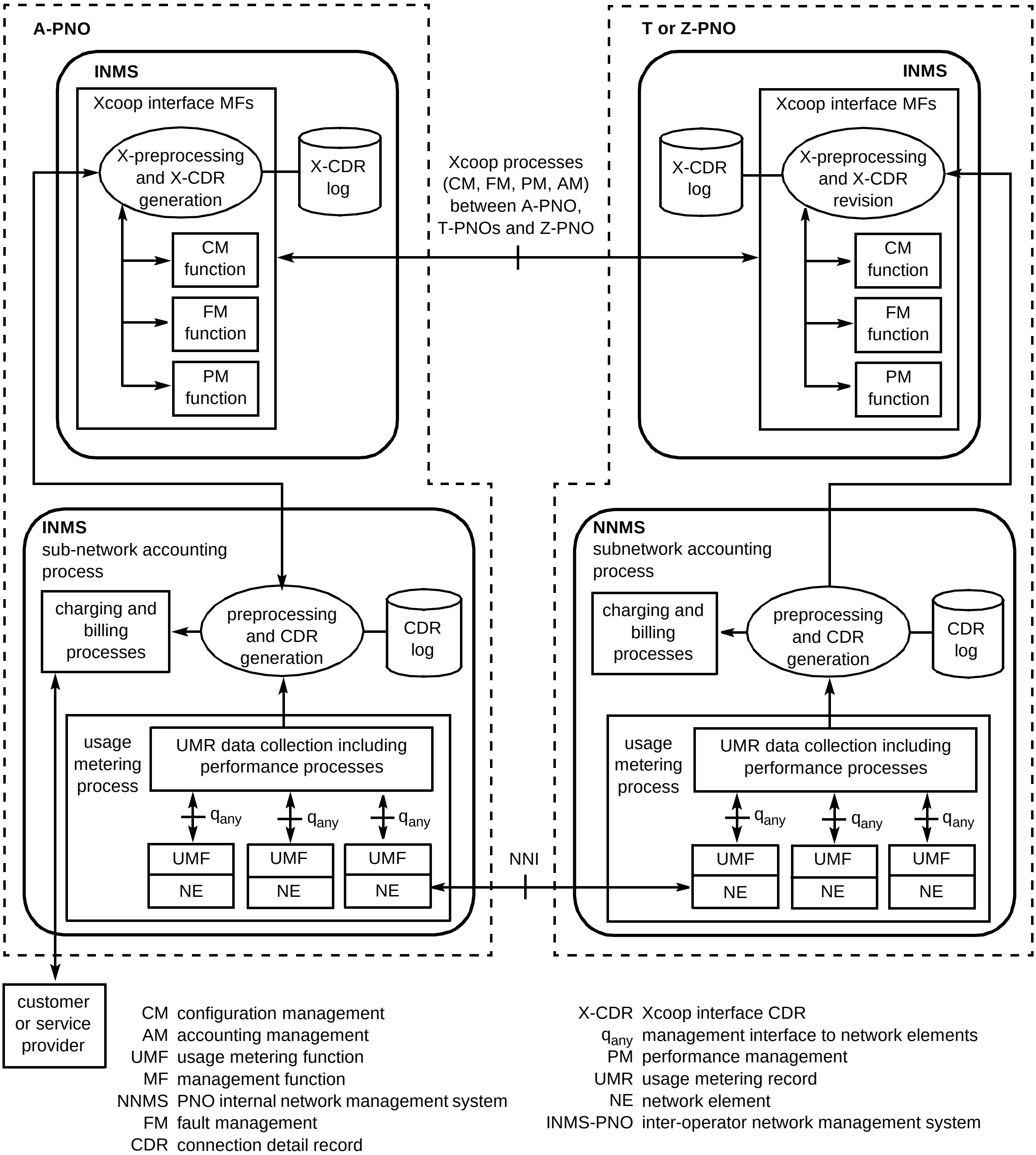


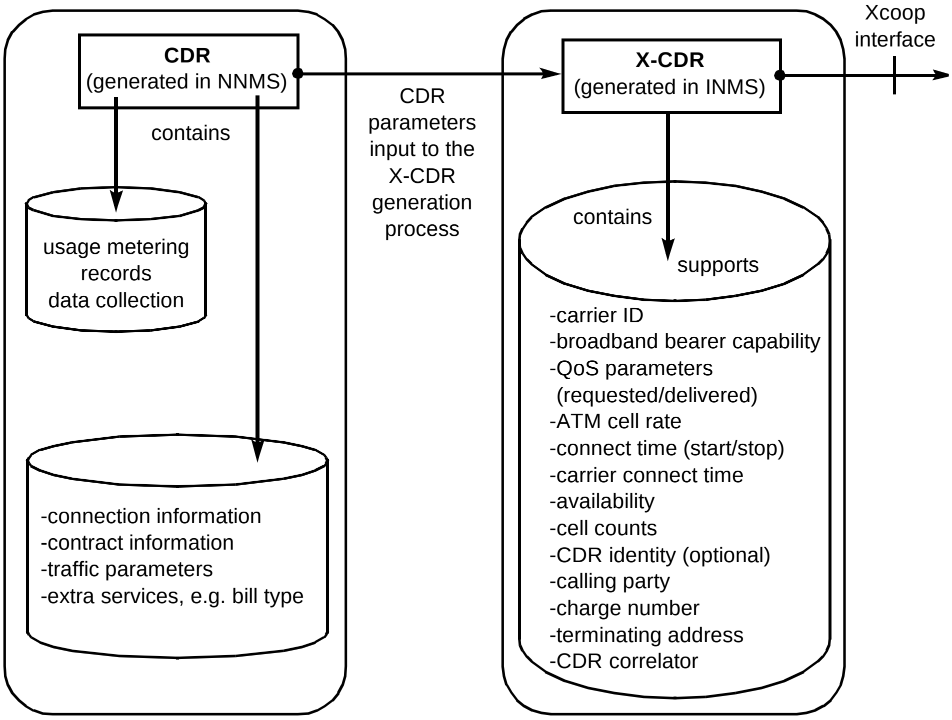
Fig 12 Reference model for accounting management using the Xcoop interface.

reasonable to postulate that creation of the X-CDR will be a function associated with the sub-network gateway, illustrated in Fig 11, as this is at the managed ingress or egress point of any PNO's sub-network domain.

The CDR data format should include information relating to the services transported and ATM packet data transfer categories (e.g. constant bit rate, variable bit rate). The CDR is transferred into the INMS part of the management system for further processing into X-CDR format and use in association with other functions of the Xcoop interface. In general, X-CDRs should contain measurement information that may be exchanged with other operators in an agreed file format.

4.5 Business and accounting management issues

Previous discussion has indicated how an Xcoop interface management function for usage-based accounting may be defined. Figure 14 shows more detail on respective service and network management aspects and especially where interactions occur to support billing, and charging



CDR connection detail record
NNMS PNO internal network management system
INMS PNO inter-operational network management system
QoS quality of service
X-CDR Xcoop interface CDR

Fig 13 Relationship between the CDR, X-CDR and Xcoop interface.

end-customers on the basis of accounting information exchanged between PNOs. Several additional management functions, likely to be required at the service management layer, are indicted for illustrative purposes but are not discussed in detail in this paper.

Revenue settlements between PNOs for bulk transfer of data over ATM networks may take place using a ‘clearing house’ scheme. This would be consistent with current practice for services such as telephony and would align with a number of ITU-T ‘D-series’ Recommendations. Such processes could be expected to be efficiently specified with usage-based accounting as the measurement basis.

5. Security for inter-operator interactions (X-interface)

This section defines the general security requirements that are considered applicable for protecting a set of defined TMN X-interface management services. A security solution is recommended, implemented and tested.

5.1 Security requirements

Whatever method is used to derive requirements, the general rule is that the security investment and the target level of protection should harmonise with the value of the assets that are at risk of being compromised. Also, indirect consequences such as bad reputation, customer dissatisfaction, legal and regulatory obligations, etc, should not be forgotten. Table 1 indicates the overall security service requirements applicable for ATM X-interface management as derived by the EURESCOM Project P708. The requirements have been defined from a simplified risk analysis based on the following four ATM management services — performance, accounting, configuration and fault.

In addition to the X-interface security requirements listed in the table, it is considered a requirement that local security services, such as security audit logging and optionally security alarm reporting, should be part of any X-interface security system.

The security requirements listed in the table are derived from the assumption that the X-interface will typically include an extensive set of ATM management capabilities. However, it is not very likely that a commercial pan-European ATM management service will have this from inception. It is more likely that initially a minimal management function set will be deployed and increased later. Nevertheless, the rest of this paper will assume that all the requirements listed above are relevant and will focus only on inter-domain (e.g. between PNOs) security protection. In addition to the above requirements, there are other characteristics that an X-interface security solution should try to fulfil:

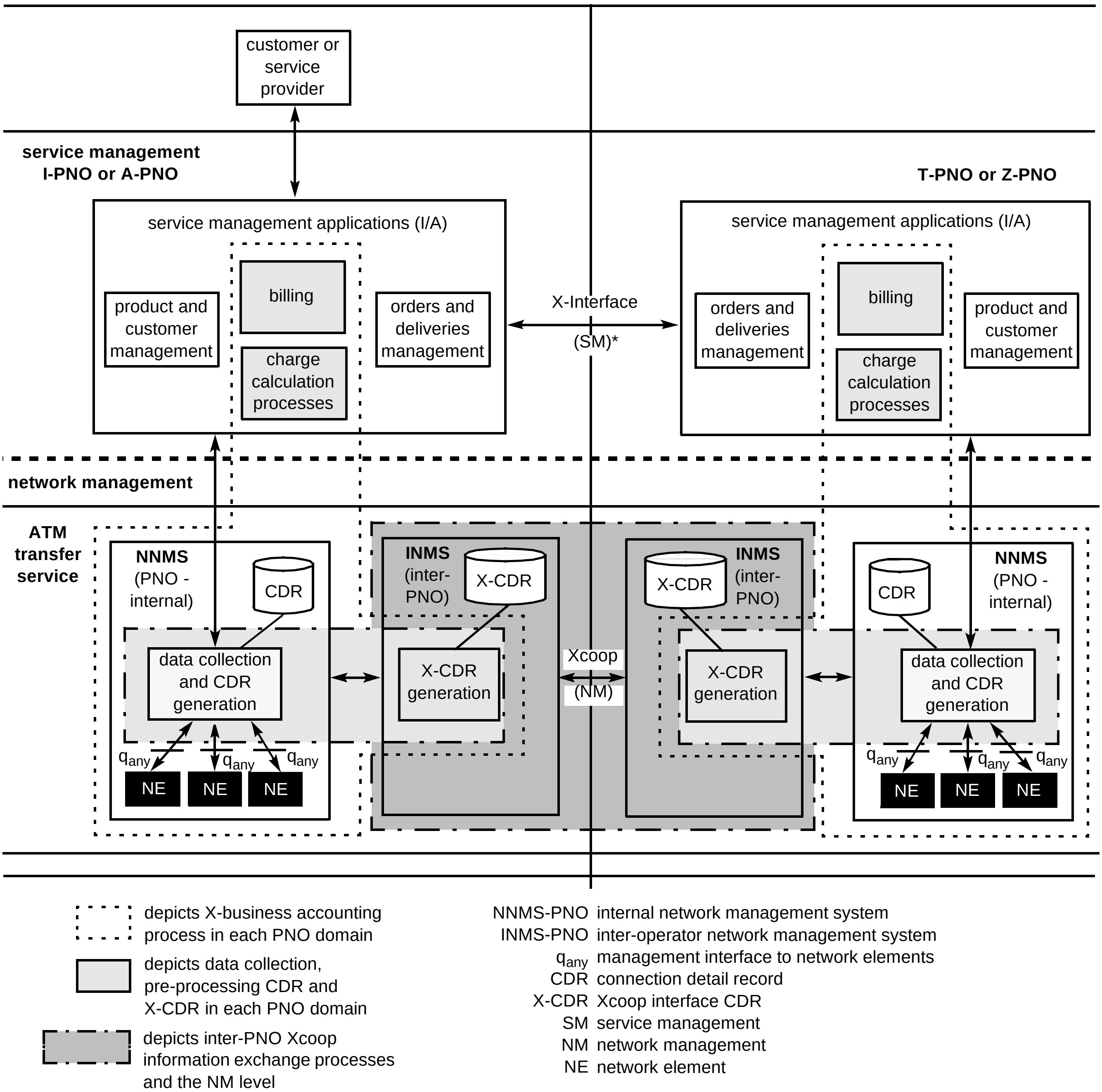


Fig 14 Xcoop business accounting process diagram.

- the solution should be deployable in a multi-vendor TMN environment,

- to the extent that end-system integration is required, it should be as simple as possible to integrate the security solution with existing management platforms and management applications,

- in order to ensure a certain level of market acceptance and off-the-shelf security product support, the security solution should preferably conform to existing security standards, including emerging TMN security standards,

- the use of standard APIs (application programming interfaces) for security services integration is preferred to facilitate easy replacement of security components and thereby a certain degree of security vendor independence,

- the solution should promote the reuse of already existing management platform security capabilities such as access control.

5.2 *The security solution*

Within a common management information protocol CMIP [9] communications environment, security services that involve cryptographic transformations upon protocol

Table 1 Security service requirements for ATM management.

X-interface security services	Requirement level	Comment
Peer-to-peer authentication	High	Mutual authentication at the granularity of individual application or application entities is required.
Access control at association establishment (Only incoming access control is addressed)	High	Access control list scheme based on authenticated calling entity is preferred. Only incoming access control is addressed.
Access control for requested management operations (Only incoming access control is addressed)	High	Access control list scheme with supporting target granularity at the level of individual managed objects. Only incoming access control is addressed.
Access control for notifications (Only incoming access control is addressed)	Low	Notification should be accepted as long as they come from a known authenticated party.
Data origin authentication	Medium	The purpose of this service is to ensure prolonged authenticity of the communicating applications after association establishment has taken place. A full integrity service at application level will serve the same purpose.
Data confidentiality protection	High	
Data confidentiality protection	High	
Non-repudiation with proof of origin	Medium	Refer to STASE-ROSE [16].
Non-repudiation with proof of delivery	Medium/High	Refer to STASE-ROSE [16].

data units (PDUs) are not particularly suited. Full confidentiality protection appears to be difficult to implement at the application level due to the fact that several CMIP parameters need to carry information in a special predefined format, a format that can no longer be guaranteed when the information is encrypted. Other security services such as integrity, data origin authentication and non-repudiation can theoretically be implemented at the application level. However, their implementation would require a rather inefficient and proprietary solution using the CMIP access control field to transfer the necessary security information. This solution would not be able to support the protection of management notifications since they do not contain the necessary CMIP protocol field. Therefore, in order to provide an effective security solution, all services that require cryptographic transformations upon CMIP messages should preferably be integrated with the communications protocol.

It is recommended that a proxy virtual private network (VPN) solution be combined with a complementary set of end-system application-level security services. Thus integrity and confidentiality services are provided for the application at the network and transport layer, over a data communications network (DCN), as depicted in Fig 15.

The proxy VPN solution will ensure integrity and confidentiality protection of exchanged management information in addition to network level authentication and access control. The application-level services will cover end-to-end authentication at the granularity of individual application entities including the necessary application-level access control services.

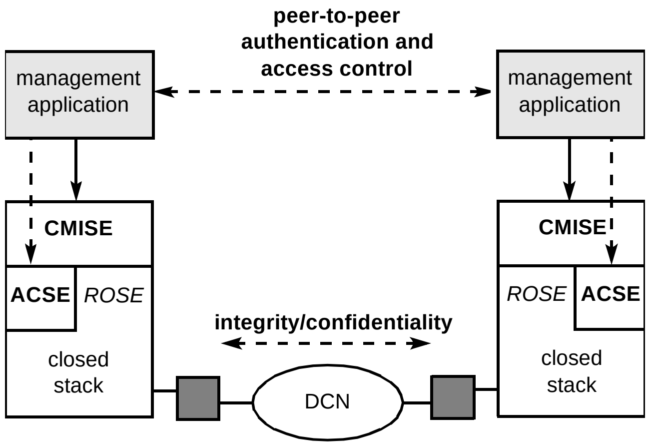


Fig 15 Overview of recommended X-interface security solution

The depicted solution does not provide an end-to-end data origin authentication at the application level — only the network path between the communicating VPN proxies is fully protected. The network connection between the VPN proxy and the end-system remains unprotected and may therefore be vulnerable to attack. The risk involved depends on what kind of network connection is used. For example, the risk would be higher if a LAN connection were used, compared to the minimum risk solution where a point-to-point cable is used. If the risk is considered high enough, one possibility to consider would be to extend the application-level solution by including a signed security token with every management message that is sent (apart from event reports). This would provide application-level data origin authentication similar to the service that is used in the US electronic bonding solution [17].

The only security service that is not supported at all by the depicted architecture is non-repudiation. The only way to get a good non-repudiation service for CMIP is through the use of Security Transformations Application Service Element for Remote Operations Service Element (STASE-ROSE) [16]. However, STASE-ROSE requires vendor support and needs stack integration.

While most X-Interface security services are aimed to protect co-operating parties from potential threats arising from external non-authorised parties, the goal of non-repudiation is to protect the co-operating parties directly from each other. Non-repudiation provides the means of collecting proof that can be submitted to a notary (e.g. a court) as valid evidence in case of a later repudiation dispute. The proxy VPN solution based on X.25 or ISDN data communication network (DCN) has not been discussed further because this solution is provided by vendors. The application-level security (see Fig 16) is more research oriented and is discussed in detail.

The architectural components are described below.

- Management application entity (MAE)

  This represents an entity that can be separately authenticated at association establishment time. A single management application may consist of one or several MAEs and there may be one or several management applications running on the same management platform.

- Security handler (SH)

  The SH is the only architectural component that needs to be tightly integrated with the management platform and/or management application. The SH must be able to intercept and take control of all incoming and outgoing association establishment attempts. As opposed to the rest of the security architecture that may be platform independent, the SH implementation will be management platform specific.

- Security control component (SCC)

  The SCC is the ‘brain’ of the security system in the sense that its responsibility is to respond to SH invocations and to co-ordinate the interactions between the other security architecture components.

- Security audit log component (SALC)

  The SALC is responsible for security audit logging with the purpose of allowing the security officer to audit security-relevant events in connection with the establishment, release and abort of management associations. All relevant security event information will be captured by the SCC and forwarded to the SALC.

- Association access control component (AACC)

  The responsibility of the AACC is to verify that the association initiator is authorised to establish an association towards a requested target MAE. The AACC will implement an ACL-based access control scheme based on ideas from ITU-T X.741 [18].

- Generic security services component (GSSC)

  The GSSC is an off-the-shelf component that supports the GSS-API as defined by IETF [19, 20]. The GSSC will support all the cryptographic functions that are

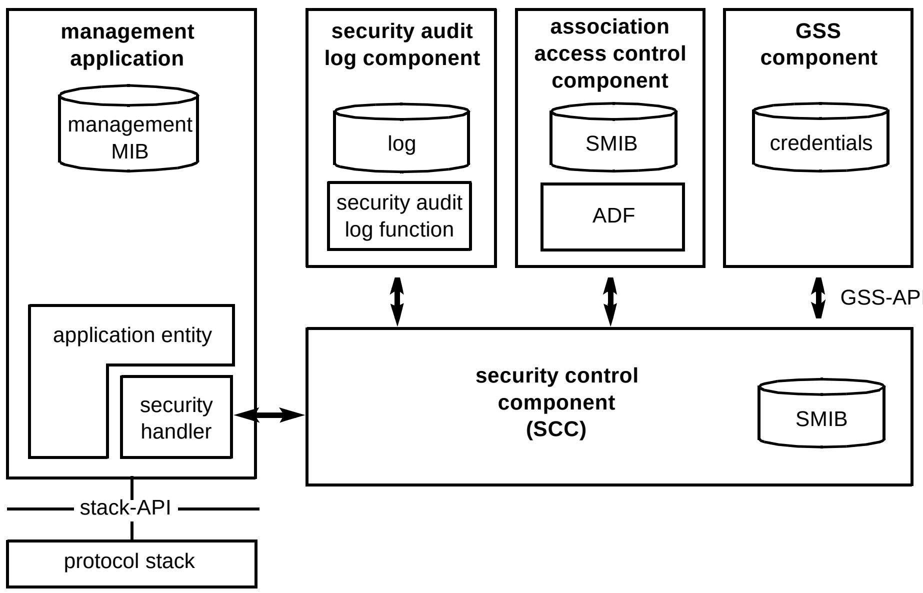


Fig 16 Application-level security architecture.

needed at the application level. The GSSC will currently only be used to support peer-to-peer authentication.

A more detailed implementation specification for the depicted architecture has been defined by the EURESCOM Project P710 deliverable [21].

5.3 Implementing and testing security for the X-interface

Work is in progress to prototype, deploy and validate the described security solution within the experimental TMN laboratory environments of the EURESCOM shareholder companies

Apart from the GSSC component that is obtained commercially off-the-shelf, the remaining application level architecture components need to be developed from a basic level. Based on component specifications developed by the EURESCOM Project P710, a prototype implementation is now available. Apart from the platform-dependent SH component, the full architecture has been implemented in a TMN platform-independent manner. Further, separate SH components have been developed for the following two TMN platforms — HP/Vertel and HP OpenView. Entrust Technologies has provided the GSSC component and related key management system. Since commercial TMN platforms have been used, the entire security architecture had to be integrated on top of the communications API provided by each individual platform. A prerequisite for this solution to be able to function is that the association establishment process can be controlled by the SH through this API. Automated association procedures, which are transparent to the application level, cannot be used in this case. Another requirement is that the API must make available an applicable ACSE/CMIP parameter that can be used to transfer the authentication tokens at association establishment time. The preferred solution would be to use the authentication parameters supported by ACSE for this purpose. However, it became apparent that these parameters were not currently available at the API level for the target management platforms. As a result, the EURESCOM implementation will make use of the CMIP A-ASSOCIATE information that is conveyed in the ‘user information’ field of ACSE.

The full prototype implementation will be deployed and tested in the PET-Lab environment.

5.4 Operational Exploitation

It is evident that commercial deployment of an X-interface without a reasonable level of security will not be acceptable to telecommunications operators. The commercial TMN platforms that are available today offer very limited support for security

Interoperability is one of the major challenges for a multi-vendor X-interface environment. If one takes the application-level solution proposed in this section, the two main requirements for security interoperability are that the communicating systems support the same GSS-API mechanism and that the same protocol field for authentication token exchange is being used. As long as these technical matters can be agreed, the X-interface security behaviour during association establishment should be clearly defined.

It should be made clear that the security solution recommended here is totally in line with the ongoing TMN security standardisation efforts. As an example, GSS-API authentication is an integral part of an emerging ETSI EN (European Norm) on TMN authentication and is also likely to be part of any ITU-T Recommendations on the same topic. STASE-ROSE [16], for example, already supports the use of GSS-API.

However, it will be impossible to be able to utilise STASE-ROSE in a multi-vendor environment if only one vendor decides to support it. Therefore, it remains to be seen when, or even if, STASE-ROSE can become a multi-vendor security solution for the X-Interface. The STASE-ROSE is a secure ROSE (as defined in the Open Systems Interconnections standards) standard which is gaining acceptance in the USA and is an ITU-T Recommendation [16].

6. Conclusions

Conclusions drawn from the work carried out in the EURESCOM projects and reported in this paper can be categorised under the main subject headings.

6.1 Performance management

Validation of the Xcoop performance management specification in the PET-Lab demonstrates the potential commercial strength of the solution. Accordingly the interface is considered suitable for the semi-permanent virtual path environment, and may be viable, subject to operational review and further development, for wider use. In other words, this work provides a corner stone for future development of performance management of ATM virtual channels and switched virtual paths. These studies have shown that the existing international standards may be inadequate for operational purposes. Further study into the practical use of performance management parameters and methods of measurement is required.

6.2 Accounting management

This work has strengthened the common industry view that availability of ATM usage-based accounting, in a co-operative environment, is a commercial requirement. The solution proposed has been developed from the existing international standards, and shows that these standards can be built upon and applied to ATM networks. It is clear that usage-based accounting is dependent on performance management and therefore the concerns in that area need to be resolved before a complete Xcoop interface solution can be specified.

6.3 Security solution

A secure environment is necessary if management systems using automated inter-PNO network management interactions are to be deployed. The proposed solution provides secure inter-PNO authentication and management application access control. It is standards based, and could be incorporated in an immediate deployment of the X-interface. The STASE-ROSE [16] TMN protocol will provide a more robust solution for a future implementation. The software to provide application-level security solution has been developed for industry-standard TMN platforms and it is expected that this development will provide a catalyst for other vendors also to incorporate security capability

Acknowledgements

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Donald Ranasinghe joined BT in 1967, while he was an undergraduate of the University of London. During his career he has worked on the development of micowave low-noise amplifiers, silicon integrated circuits (in particular, processing, packaging, testing and quality assurance), and the design validation of prototype integrated circuits, and his pioneering team effort in the ESPRIT work program provided an industry-standard product for prototyping VLSI semi-conductors. In 1993 he moved into the Network Communications Management Research area and worked on a number of European collaborative projects. He has a degree in Electrical and Electronic Engineering and is an Associate Member of the IEE.

Colin Smith is a Chartered Engineer and has been involved in many areas of switching systems development and operational deployment over a period of 30 years, including the assessment of SystemX performance, network design and software for maintenance-related purposes. More recently he led the ITU-T group that developed, jointly with ISO, the X.700 series of Telecommunications Management Network (TMN) Interface Recommendations. He was the project leader for the EURESCOM ATM studies and also led the development of the Performance Ensemble.

Christopher Hatch graduated with a BSc (Hons) degree in Physics from Exeter University and an MSc and PhD in Quantum Electronics and Physics respectively in 1974 and 1977 from Essex University. After a period working for Philips Research Labs on novel flat-screen display technology, he joined BT Laboratories in 1980. His responsibilities included the development of non-destructive optical analysis techniques for semiconductor device wafers. Since 1994 he has participated in the Communications Management Research Campaign and has been a BT representative in several European collaborative projects within the EURESCOM, ETNO and ETSI organisations. He is a Chartered Physicist and a Member of the Institute of Physics.

Stephen Beaumont joined BT in 1986 after graduating from Strathclyde University with a BSc/BEng in Electrical and Electronic Engineering. He has worked on video systems, ATM and packet video, a facial recognition demonstrator, and the management systems of the BT Touchpoint mulitmedia kiosk project.

In 1997 he joined the Communications Management Research campaign, working in the Network Management Research group, where he has been the team leader in the EURESCOM P708 project ‘TMN X-Interface Studies and Experiments for ATM’.

Pal Kristiansen graduated from the Norwegian Institute of Technology in 1993 and has been working at the Telenor Research and Development in the security area, since 1994. He has specialised in the field of security for TMN since joining Telenor R&D at Kjeller, Norway and is an active participant of ETSI. He has participated in several EURESCOM projects and held the position of the project leadership for the EURESCOM P710 project. He is currently working on the security aspects such as PKI (Public Key Infrastructure) and TTP (Trusted Third Party) for telecommunications applications.